

Problem

A parallel resonant circuit with a bandwidth of 40 krad/s and the half-power frequencies are $\omega_1 = 4.98$ Mrad/s and $\omega_2 = 5.02$ Mrad/s, calculate the quality factor and resonant frequency.

Step-by-step solution

Step 1 of 2

Write the formula for half power frequency.

$$\omega_1 = \omega_0 - \frac{B}{2}$$

Substitute 4.98 Mrad/s for ω_1 and 40 krad/s for B .

$$4.98 \times 10^6 = \omega_0 - \frac{40 \times 10^3}{2}$$

$$\omega_0 = 4.98 \times 10^6 + 20 \times 10^3$$

$$\omega_0 = 5 \text{ Mrad/s}$$

Therefore, the resonant frequency, ω_0 is **5 Mrad/s**.

Step 2 of 2

Calculate the quality factor using the following formula.

$$B = \frac{\omega_0}{Q}$$

$$Q = \frac{\omega_0}{B}$$

$$= \frac{5 \times 10^6}{40 \times 10^3}$$

$$= 125$$

Therefore, the quality factor of the circuit, Q is **125**.